

Machine Learning (机器学习)

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网安C2-1025



Curriculum assessment

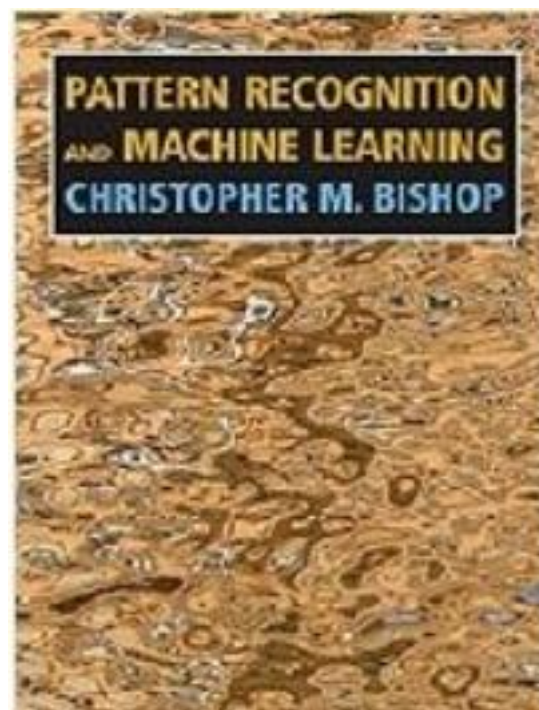
- ✎ Exam(70%) : final exam
- ✎ Practice(15%): Homework+Experiment+attendance
- ✎ One Project report(15%): **Teamwork**
 - ✎ State-of-art results: 100%

Submit via E-learning.

References



- Tom M. Mitchell, 机器学习.原书名: *MACHINE LEARNING* (英文版), 机械工业出版社, 2003-3-1.

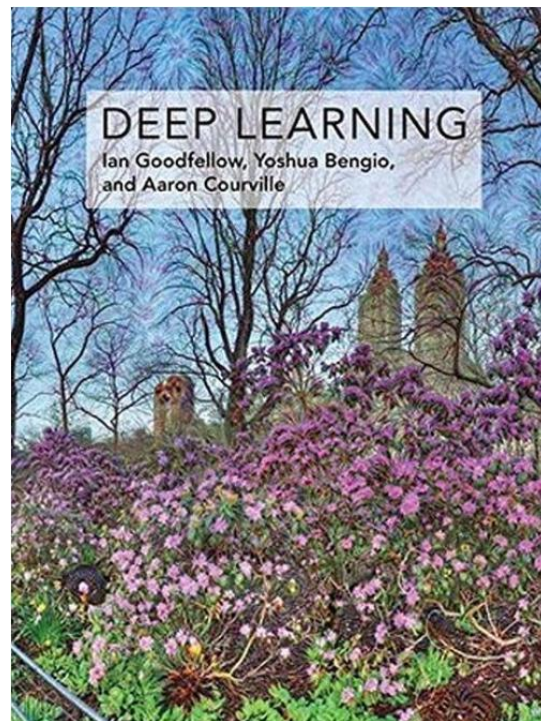


- Christopher M Bishop, 《*Pattern Recognition and Machine Learning*》, 2006.

References



周志华,《机器学习》,清华大学出版社,2016.



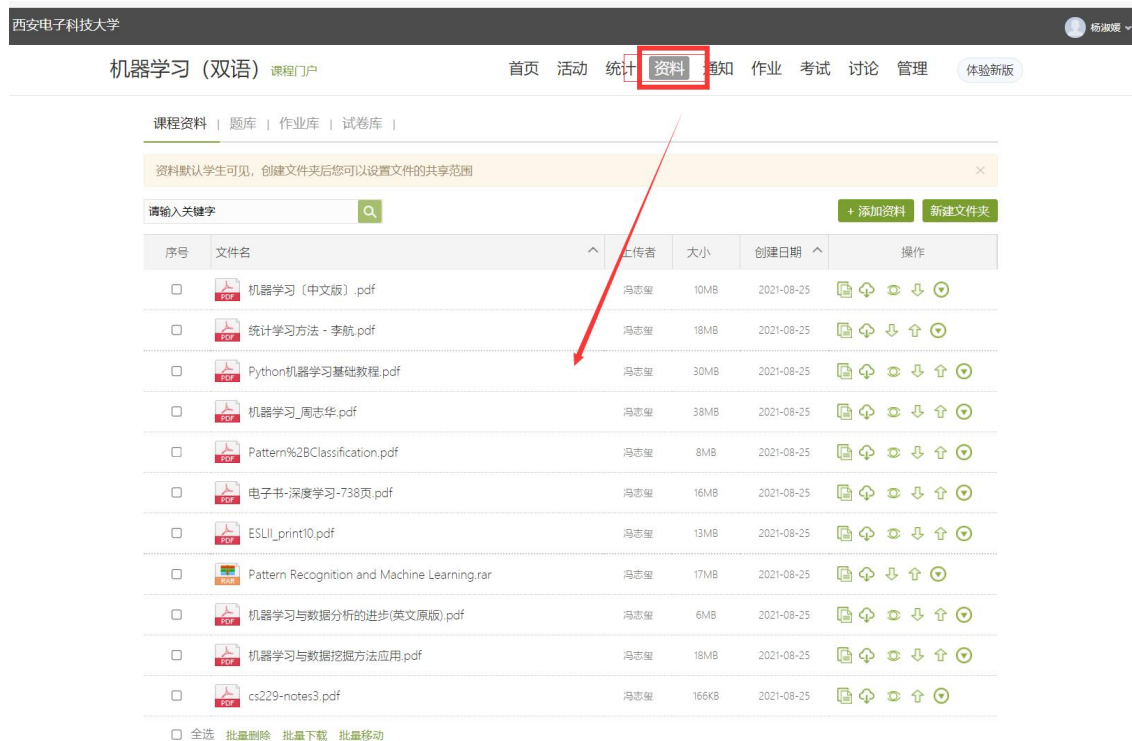
Ian Goodfellow, Yoshua Bengio and Aaron Courville, Deep Learning, 2016, MIT Press.

<http://www.deeplearningbook.org>

References

- all the reference could be download from E-learning of our course.

<http://mooc1.chaoxing.com/course/219032799.html>



The screenshot displays the Xue301 E-learning platform interface. At the top, the header includes the university name '西安电子科技大学' (Xi'an Jiaotong University) and a navigation bar with links: '机器学习 (双语)' (Machine Learning (Bilingual)), '课程门户' (Course Portal), '首页' (Home), '活动' (Activities), '统计' (Statistics), '资料' (Materials), '通知' (Notifications), '作业' (Assignments), '考试' (Exams), '讨论' (Discussions), and '管理' (Management). The '资料' (Materials) link is highlighted with a red box. Below the navigation bar, there is a search bar with the placeholder text '请输入关键字' (Please enter keywords) and a search icon. To the right of the search bar are two buttons: '+ 添加资料' (Add Materials) and '新建文件夹' (New Folder). The main content area displays a table of course materials. The table has columns for '序号' (Serial Number), '文件名' (File Name), '上传者' (Uploader), '大小' (Size), '创建日期' (Creation Date), and '操作' (Operations). A red arrow points to the '资料' (Materials) link in the navigation bar. The table lists various PDF files related to machine learning, including '机器学习 [中文版].pdf', '统计学习方法 - 李航.pdf', 'Python机器学习基础教程.pdf', '机器学习_周志华.pdf', 'Pattern%2BClassification.pdf', '电子书-深度学习-738页.pdf', 'ESLII_print10.pdf', 'Pattern Recognition and Machine Learning.rar', '机器学习与数据分析的进步(英文原版).pdf', '机器学习与数据挖掘方法应用.pdf', and 'cs229-notes3.pdf'. At the bottom of the table, there are links for '全选' (Select All), '批量删除' (Batch Delete), '批量下载' (Batch Download), and '批量移动' (Batch Move).

序号	文件名	上传者	大小	创建日期	操作
☐	机器学习 [中文版].pdf	冯志军	10MB	2021-08-25	    
☐	统计学习方法 - 李航.pdf	冯志军	18MB	2021-08-25	    
☐	Python机器学习基础教程.pdf	冯志军	30MB	2021-08-25	    
☐	机器学习_周志华.pdf	冯志军	38MB	2021-08-25	    
☐	Pattern%2BClassification.pdf	冯志军	8MB	2021-08-25	    
☐	电子书-深度学习-738页.pdf	冯志军	16MB	2021-08-25	    
☐	ESLII_print10.pdf	冯志军	13MB	2021-08-25	    
☐	Pattern Recognition and Machine Learning.rar	冯志军	17MB	2021-08-25	    
☐	机器学习与数据分析的进步(英文原版).pdf	冯志军	6MB	2021-08-25	    
☐	机器学习与数据挖掘方法应用.pdf	冯志军	18MB	2021-08-25	    
☐	cs229-notes3.pdf	冯志军	166KB	2021-08-25	    

☐ 全选 [批量删除](#) [批量下载](#) [批量移动](#)

Definition of Learning

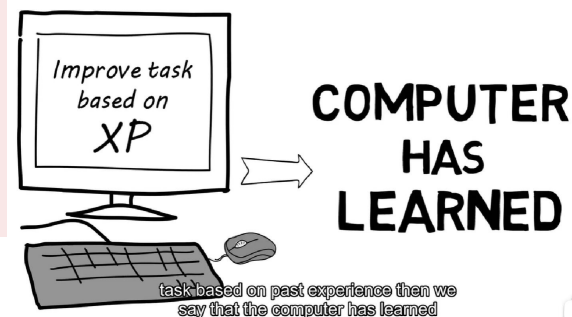
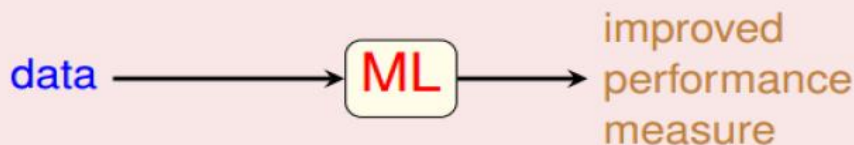
➤ “学习” --- 赫尔伯特·西蒙 (Herbert A. Simon)

“Learning is any process by which a system improves performance from experience.”

机器学习就是计算机系统通过运用数据及统计方法提高系统性能的机器学习



machine learning: improving some performance measure with experience **computed** from **data**



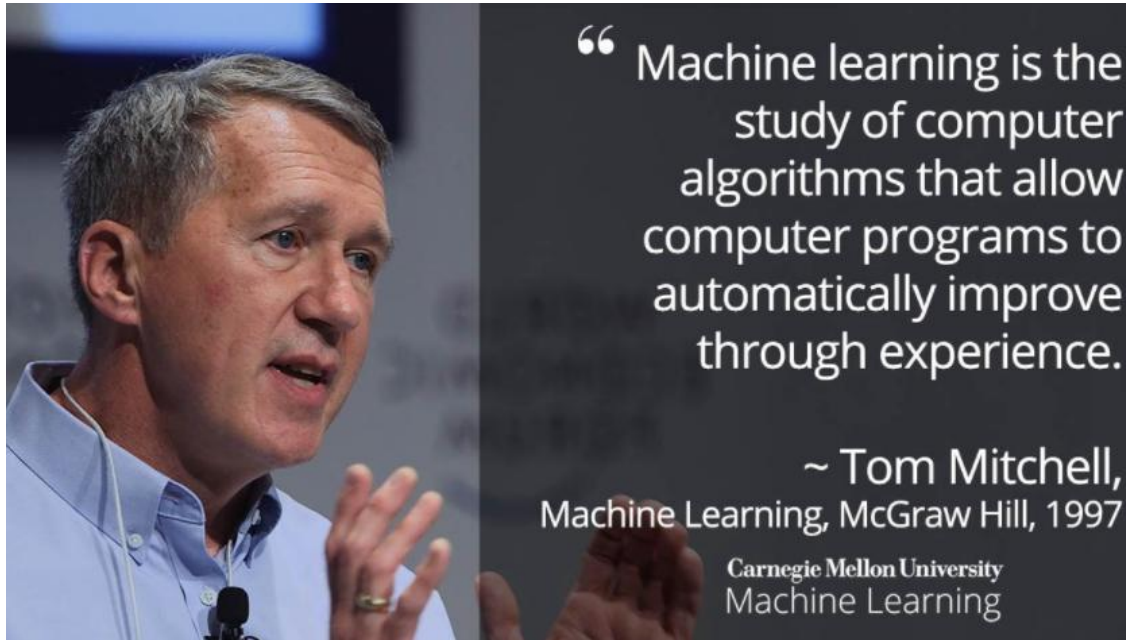
Definition of Learning

Definition by Tom Mitchell (1998):

Machine Learning is the study of algorithms that

- improve their performance P
- at some task T
- with experience E .

A well-defined learning task is given by $\langle P, T, E \rangle$.



“ Machine learning is the study of computer algorithms that allow computer programs to automatically improve through experience.

~ Tom Mitchell,
Machine Learning, McGraw Hill, 1997

Carnegie Mellon University
Machine Learning

Contents

- What is Machine Learning
- Applications of Machine Learning
- Components of Machine Learning
- Machine Learning and Other Fields

Contents

- What is Machine Learning
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From Learning to Machine Learning

learning: acquiring **skill**
with experience accumulated from **observations**



machine learning: acquiring **skill**
with experience accumulated/**computed** from **data**



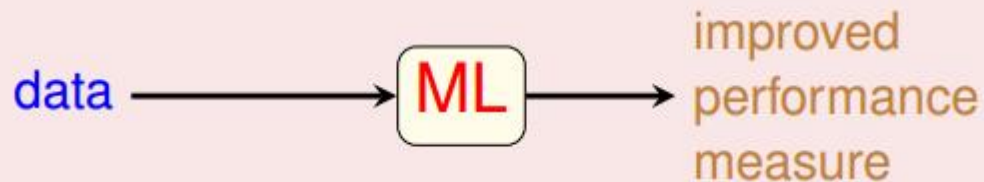
- what is **skill** ?

what is skill ?

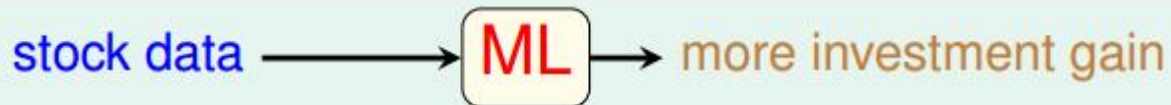
skill

⇔ improve some performance measure (e.g. prediction accuracy)

machine learning: improving some performance measure
with experience **computed** from data

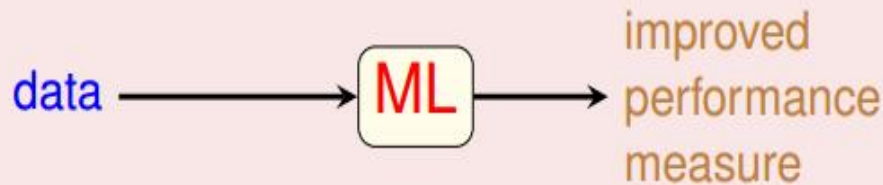


An Application in Computational Finance



Key Essence of Machine Learning

machine learning: improving some performance measure
with experience **computed** from data



- 1 exists some 'underlying pattern' to be learned
—so 'performance measure' can be improved
- 2 but **no** programmable (easy) **definition**
—so 'ML' is needed
- 3 somehow there is **data** about the pattern
—so ML has some 'inputs' to learn from

Machine Learning Route

ML: an **alternative route** to build complicated systems

Some Use Scenarios

- when human cannot program the system manually
—navigating on Mars
- when human cannot 'define the solution' easily
—speech/visual recognition
- when needing rapid decisions that humans cannot do
—high-frequency trading
- when needing to be user-oriented in a massive scale
—consumer-targeted marketing

Definition of Learning

Traditional Programming



Machine Learning



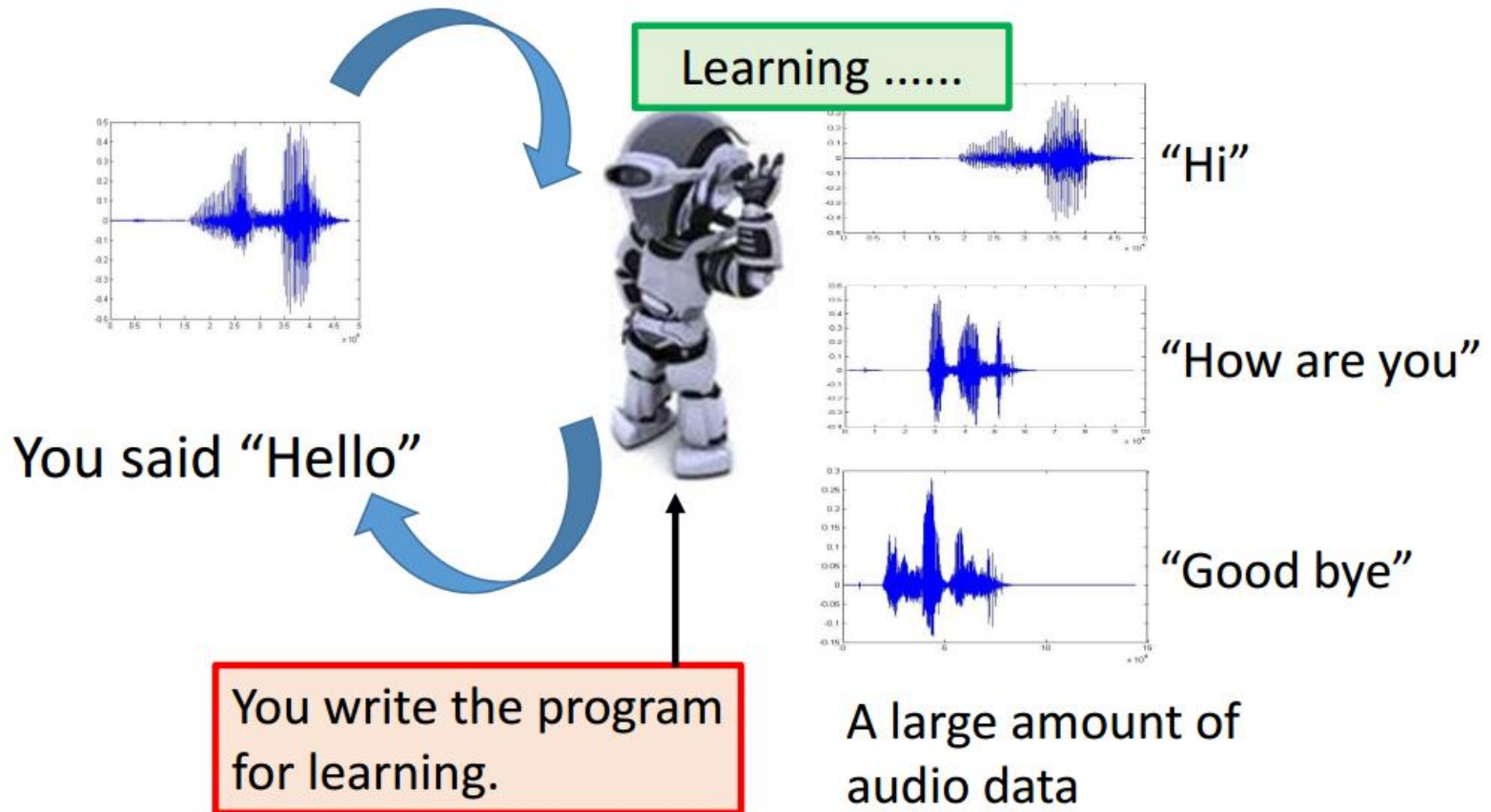
Q

- **Which of the following is best suited for machine learning?**
-
- A. Predicting whether the next cry of the baby girl happens at an even-numbered minutes or not
- B. Determining whether a given graph contains a cycle
- C. Deciding whether to approve the credit card to some customer
- D. Guessing whether the earth will be destroyed by the misuse of nuclear power in the next ten years

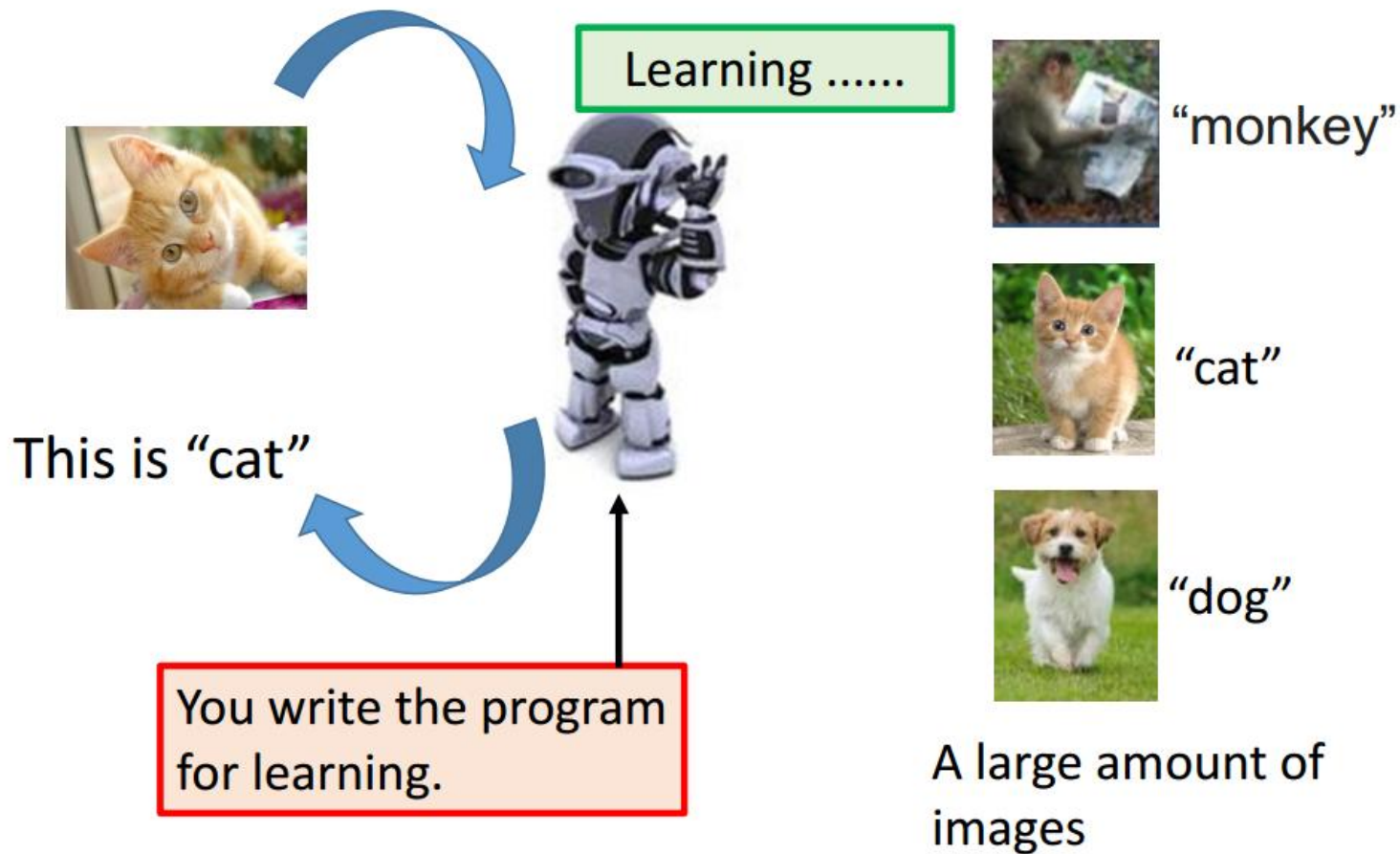
Contents

- What is Machine Learning
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Some Examples



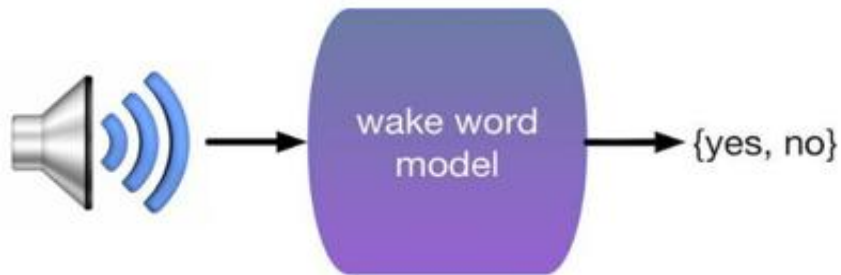
Some Examples



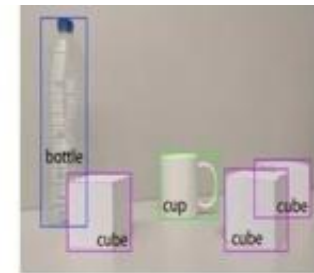
Some Examples



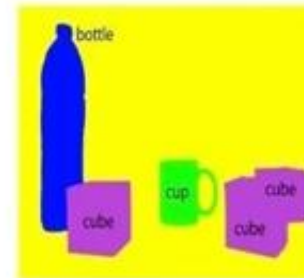
猫 猫 汪 汪



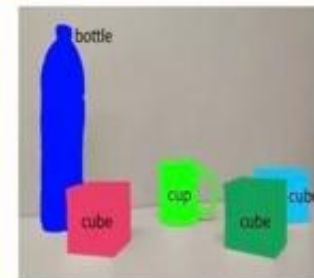
(a) Image classification



(b) Object localization



(c) Semantic segmentation



(d) Instance segmentation



A woman is throwing a **frisbee** in a park.

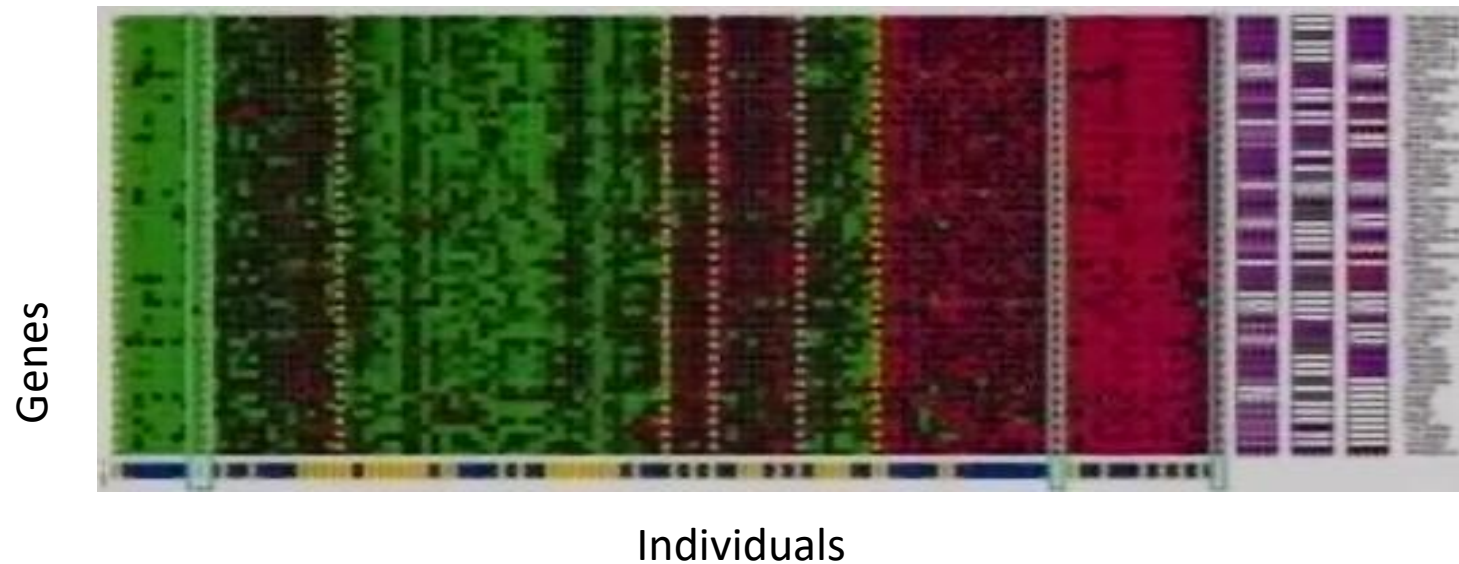


A **dog** is standing on a hardwood floor.



A **stop** sign is on a road with a mountain in the background

Some Examples

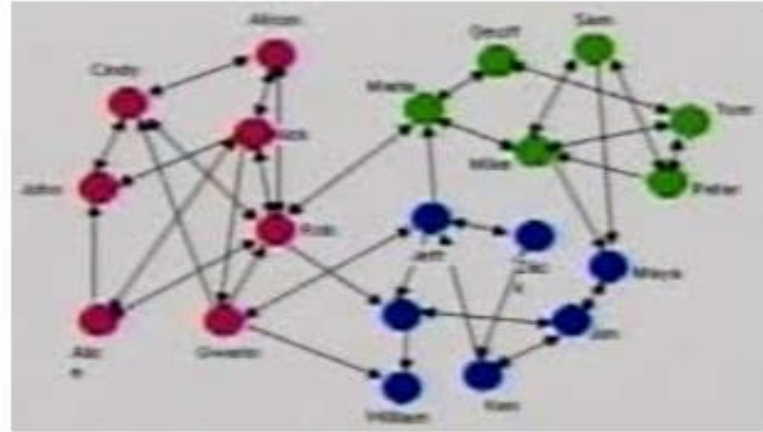


This is an example applying unsupervised learning algorithms to try to understand gene data. It is trying to look at genes as individuals and group them into clusters based on properties of what genes they respond to — based on properties of how the genes respond to different experiments.

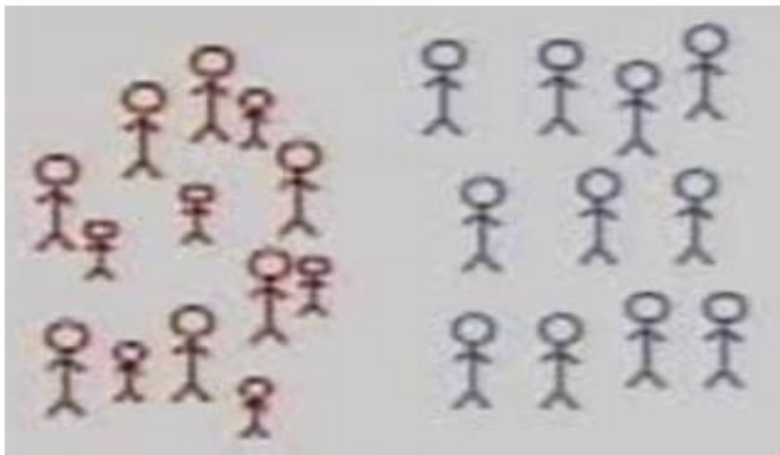
Some Examples



Organizing computing clusters



Social network analysis



Market segmentation



Astronomical data analysis

Some Examples



a robot dog



a robotic snake



A car getting over an obstacle

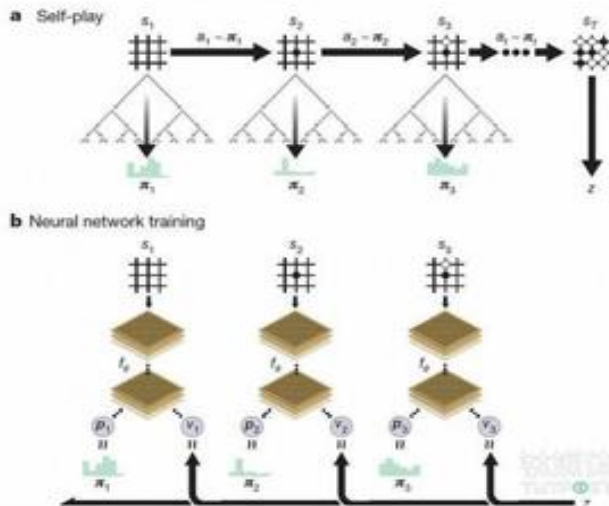
Some Examples



October 5 – 9, 2015

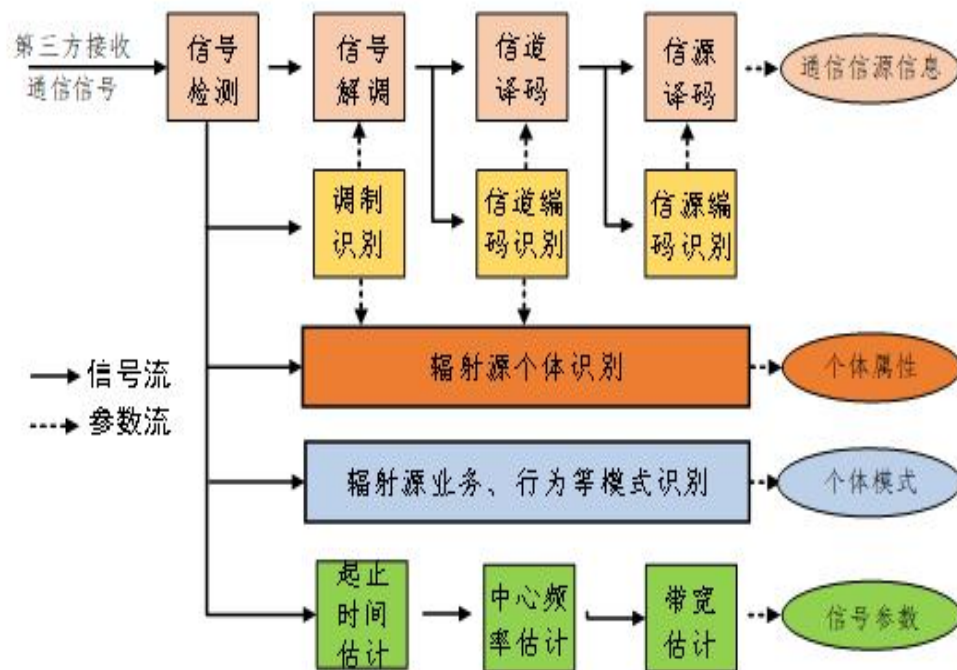
<Official match>

- Time limit: 1 hour
- AlphaGo Wins (5:0)



Waymo公测的L4级无人驾驶汽车

Some Examples



Some Examples



When Do We Use Machine Learning?

- Daily Needs: Food, Clothing, Housing, Transportation

1 Food (Sadilek et al., 2013)

- **data:** Twitter data (words + location)
- **skill:** tell food poisoning likeliness of restaurant properly

2 Clothing (Abu-Mostafa, 2012)

- **data:** sales figures + client surveys
- **skill:** give good fashion recommendations to clients

3 Housing (Tsanas and Xifara, 2012)

- **data:** characteristics of buildings and their energy load
- **skill:** predict energy load of other buildings closely

4 Transportation (Stallkamp et al., 2012)

- **data:** some traffic sign images and meanings
- **skill:** recognize traffic signs accurately

ML is everywhere!

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Types of Learning

Machine Learning
 $\approx$ Looking for a Function

- Speech Recognition

$$f(\text{[audio waveform]}) = \text{"How are you"}$$

- Image Recognition

$$f(\text{[cat image]}) = \text{"Cat"}$$

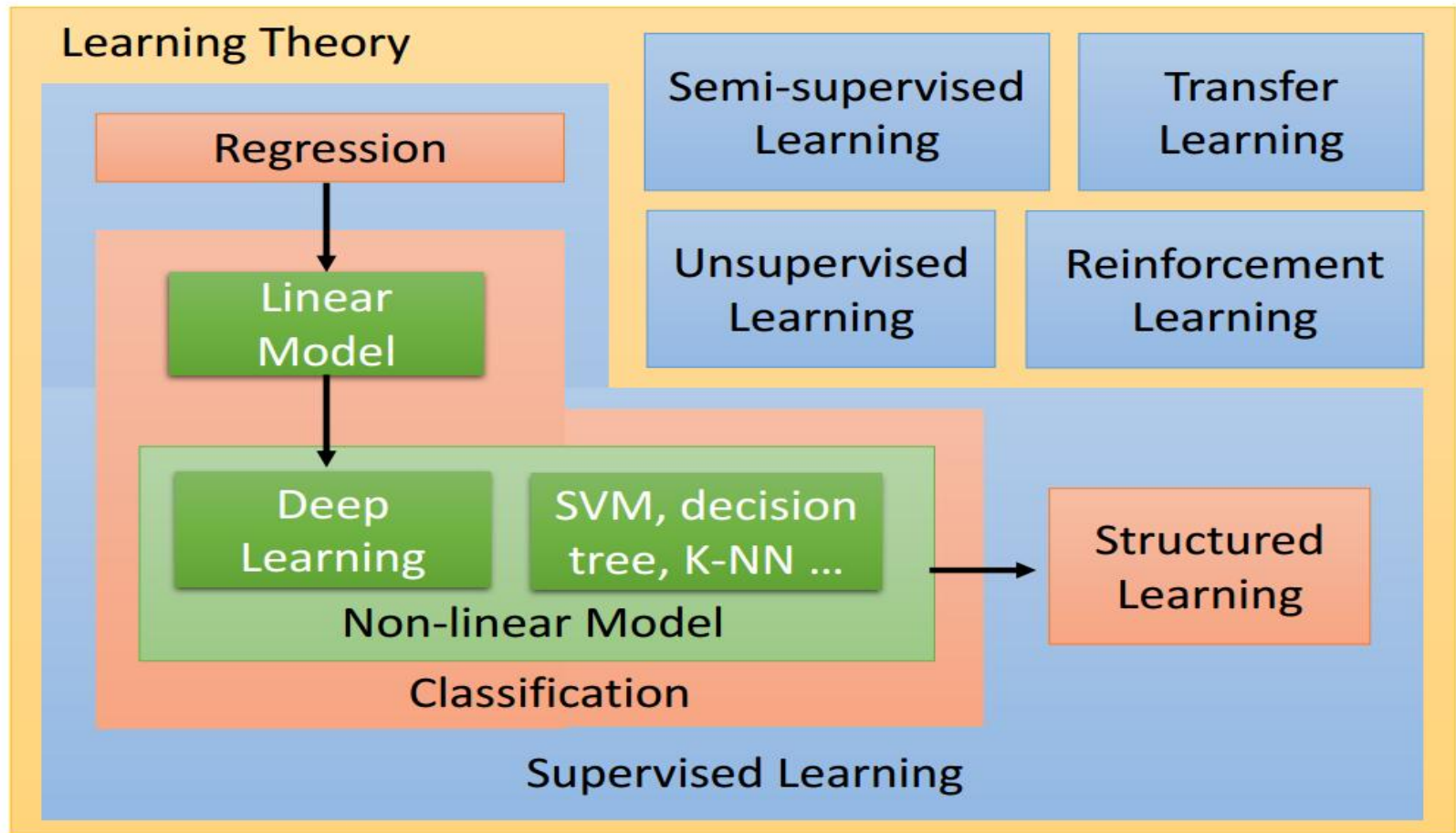
- Playing Go

$$f(\text{[Go board image]}) = \text{"5-5"}_{\text{(next move)}}$$

- Dialogue System

$$f(\text{"Hi"}_{\text{(what the user said)}}) = \text{"Hello"}_{\text{(system response)}}$$

Types of Learning



task

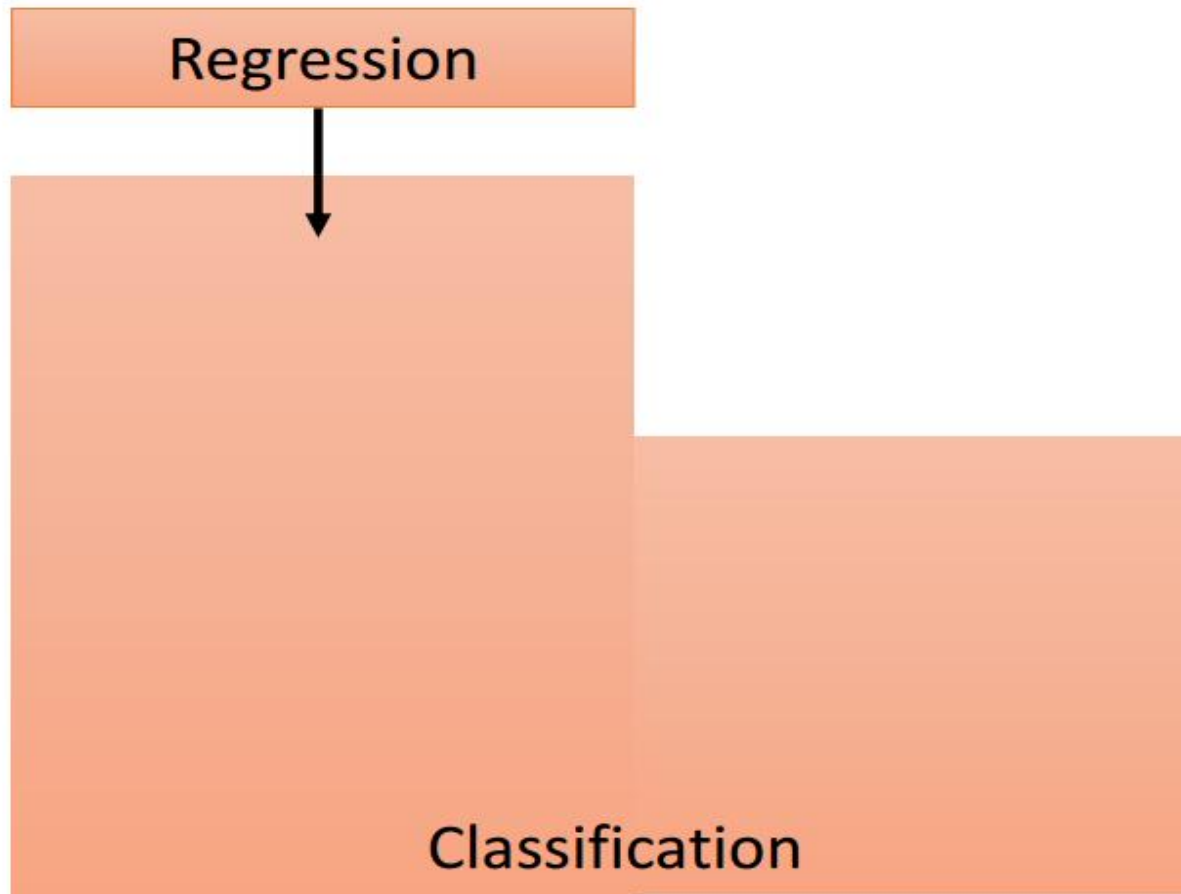


method



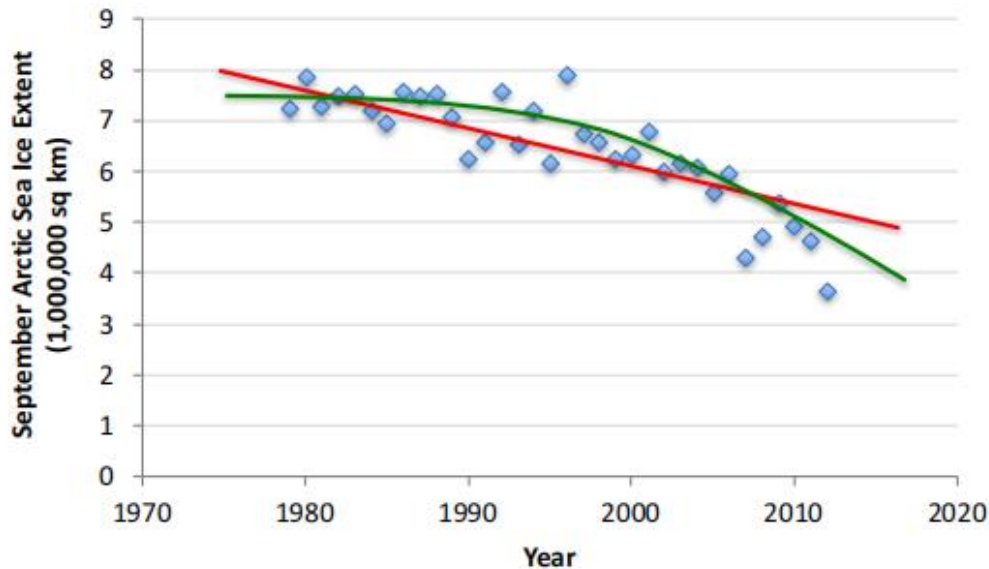
scenario

Machine Learning: (1) Task



(1) Task: Regression

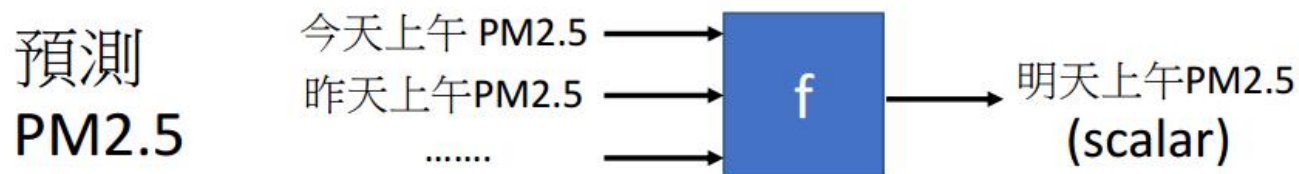
- Given $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$
- Learn a function $f(x)$ to predict y given x
 - -- y is a real-valued ==regression



Regression

Regression

The output of the target function f is “scalar”.



Training Data:

Input:

9/01 上午 PM2.5 = 63 9/02 上午 PM2.5 = 65

Input:

9/12 上午 PM2.5 = 30 9/13 上午 PM2.5 = 25

Output:

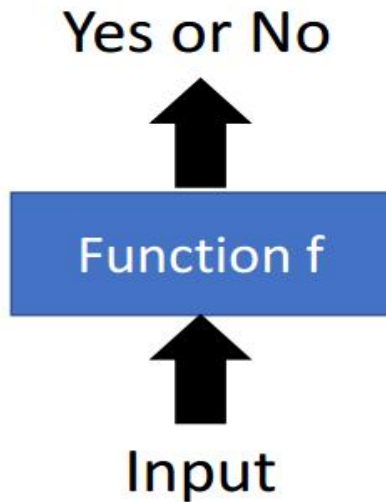
9/03 上午 PM2.5 = 100

Output:

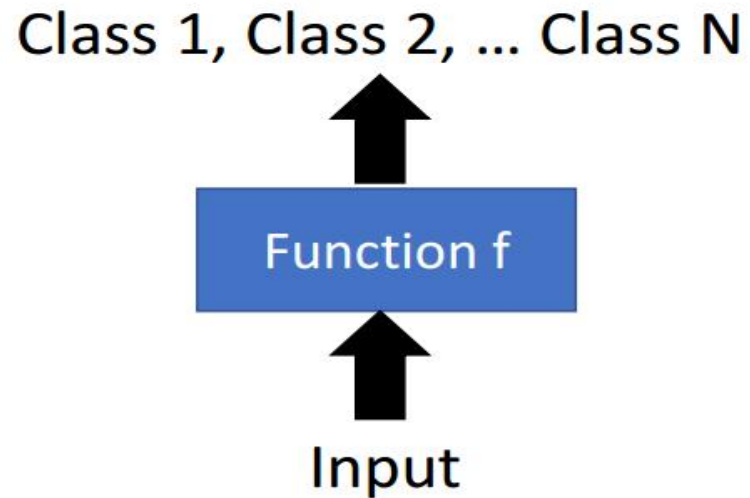
9/14 上午 PM2.5 = 20

(1) Task: Classification

- Binary Classification

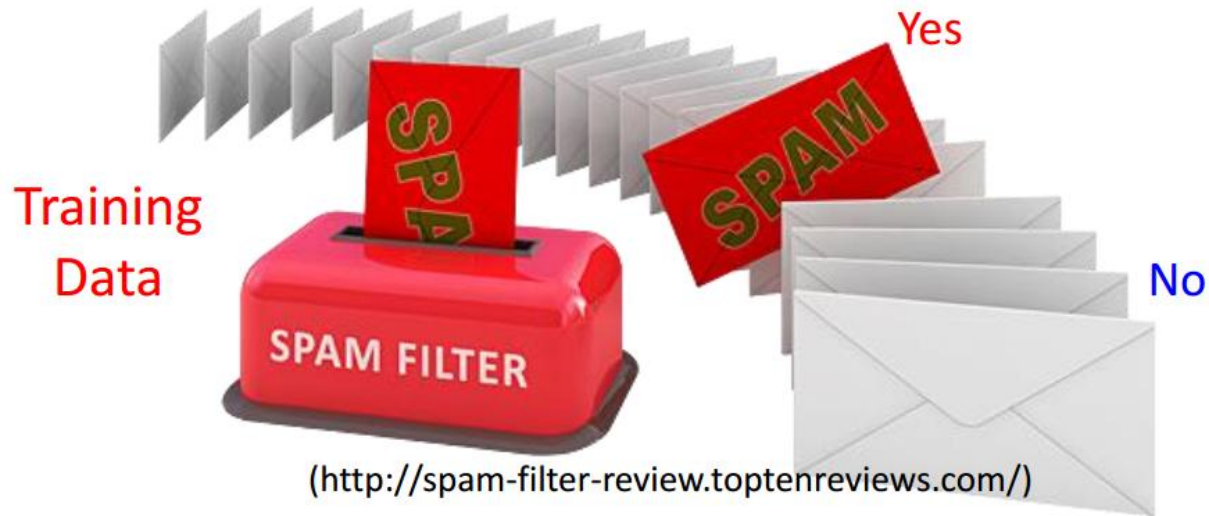


- Multi-class Classification



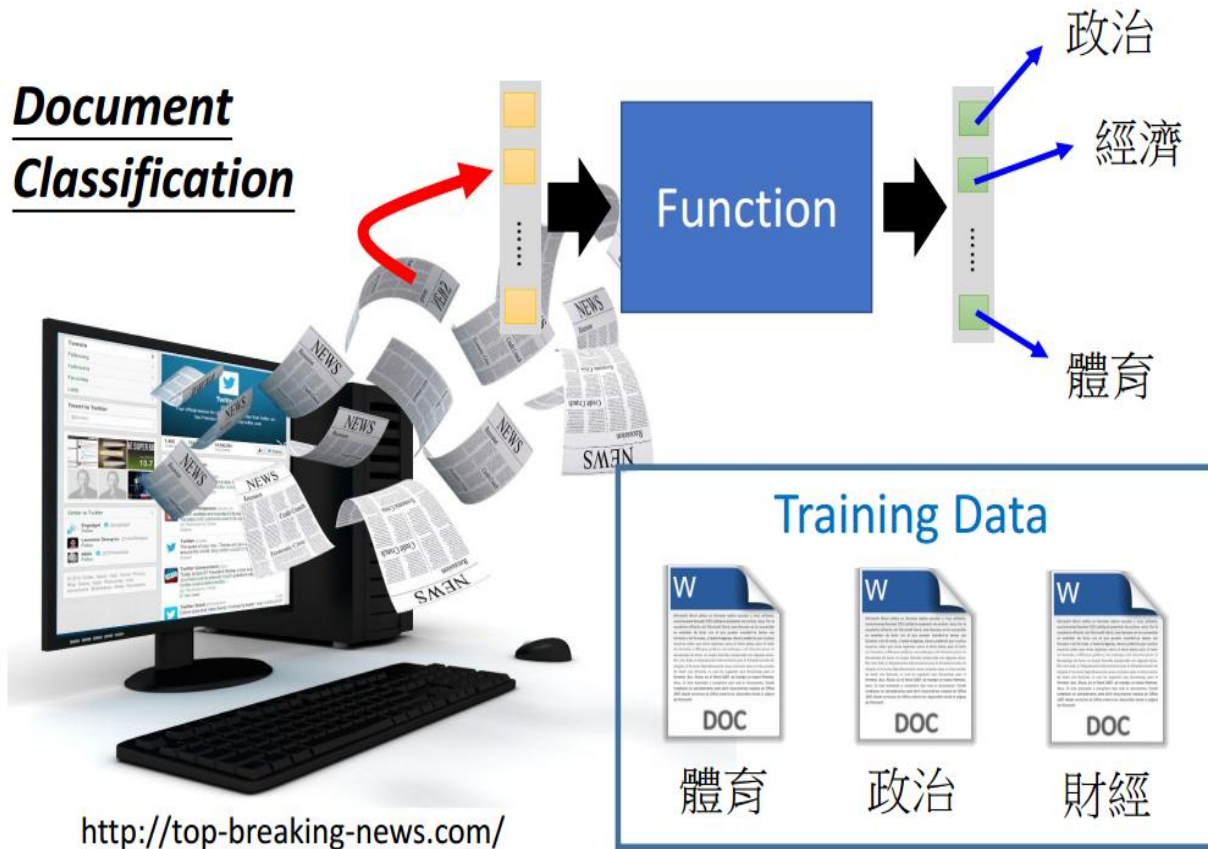
Binary Classification

Spam
filtering



Multi-class Classification

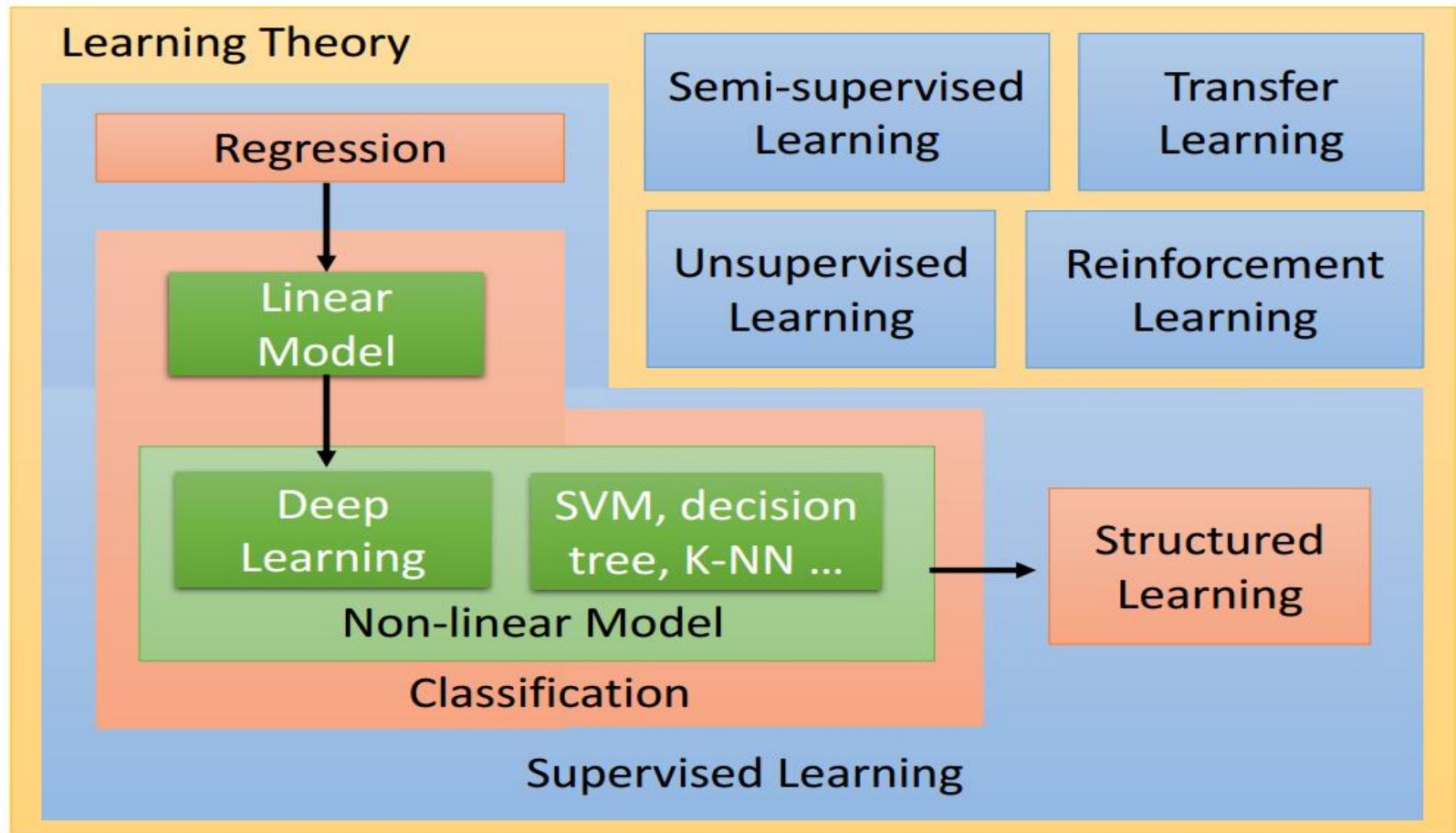
Document Classification



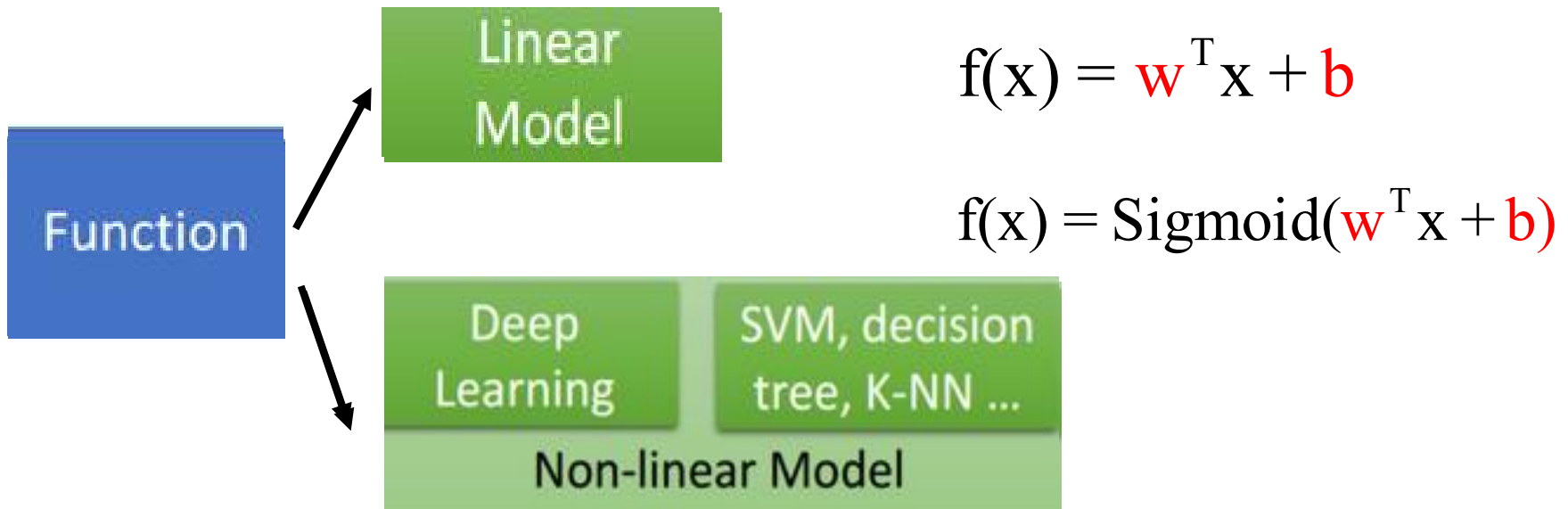
(1) Task: Structured Learning



Types of Learning

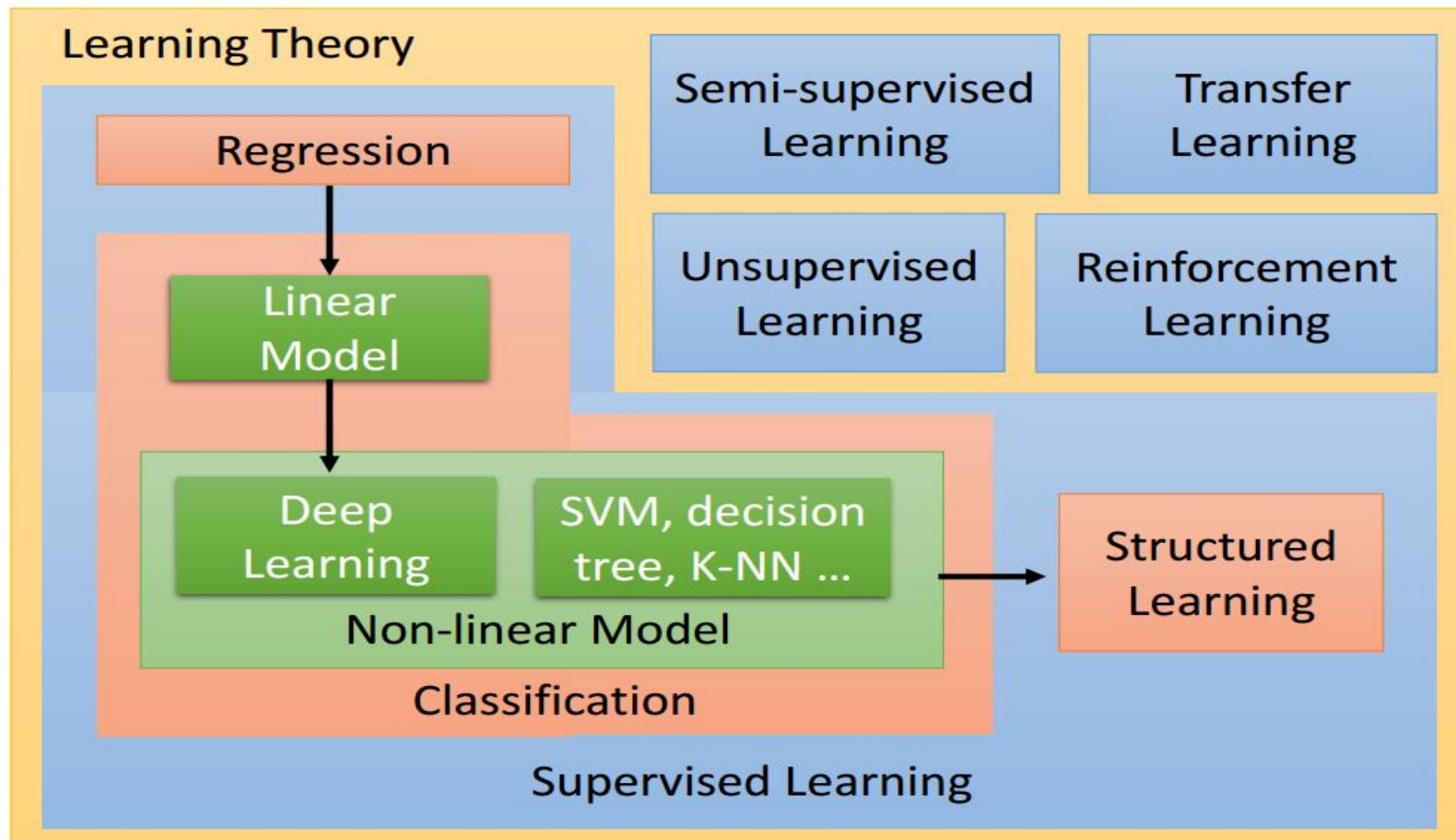


Machine Learning: (2) Method

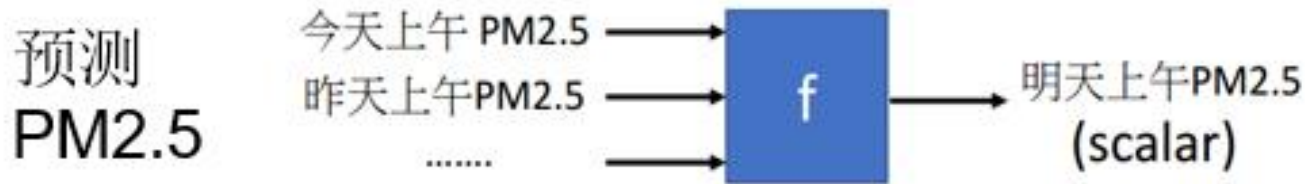


$$f(\mathbf{x}) = g^n \left(\dots g^2 \left((\mathbf{w}^2)^T g^1 \left((\mathbf{w}^1)^T \mathbf{x} + \mathbf{b}^1 \right) + \mathbf{b}^2 \right) \right)$$

Types of Learning



Machine Learning: (3) Scenario



Training Data:

Input:

9/01 上午 PM2.5 = 63 9/02 上午 PM2.5 = 65

Output:

9/03 上午 PM2.5 = 100

Input:

9/12 上午 PM2.5 = 30 9/13 上午 PM2.5 = 25

Output:

9/14 上午 PM2.5 = 20

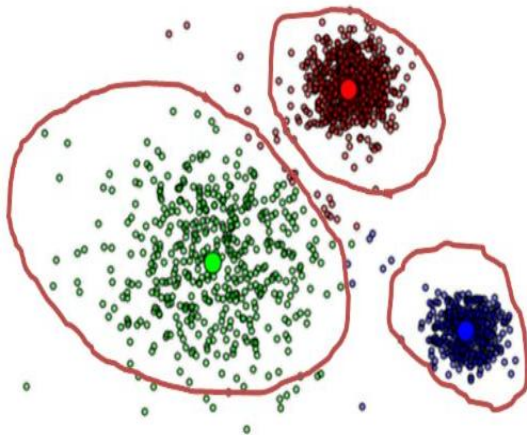
Supervised Learning

$f?$ Labeled Training Data = $\{(x_i, y_i)_{i=1}^Q\}$

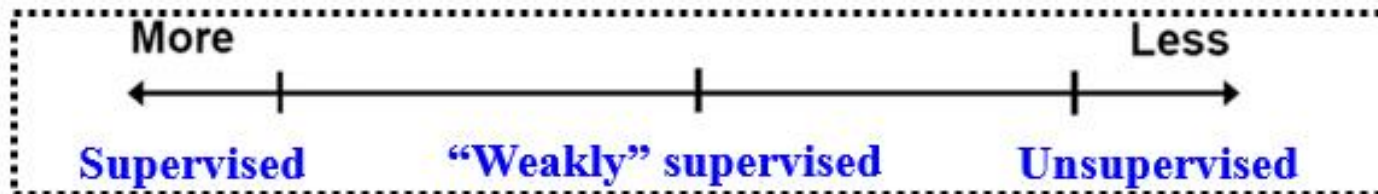
Machine Learning: (3) Scenario

Unsupervised Learning

Clustering:



For example: K-means, Agglomerative clustering, Mean-shift clustering, Spectral clustering



Machine Learning: (3) Scenario

For example, recognizing cats and dogs

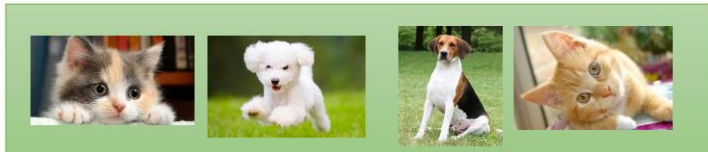
Labelled data



Weakly supervised Learning



Unlabeled data



(Images of cats and dogs)

Semi-supervised



Data not related to the task considered
(can be either labeled or unlabeled)

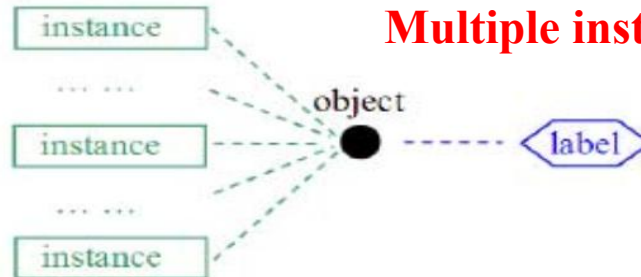
Transfer-learning

Machine Learning: (3) Scenario

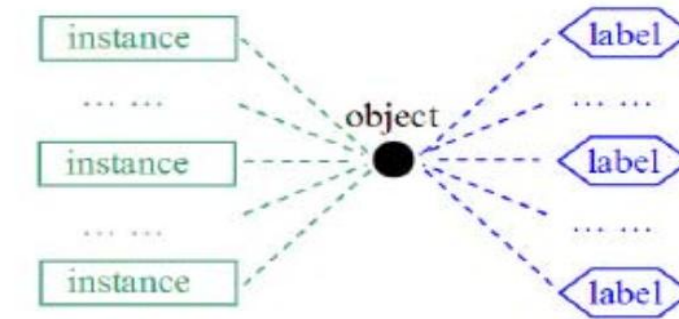
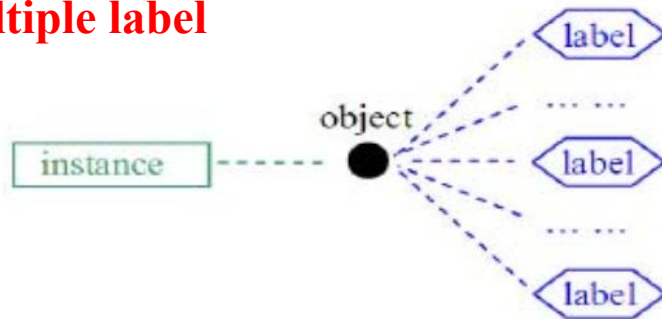
Supervised



Multiple instance



Multiple label



Multiple instance and multiple label

South Africa's FIFA World Cup – a success at home and abroad



Success, pride and unity – three words which are being used by the people of South Africa to describe the effects that staging the 2010 FIFA World Cup™ has had on their country. These feelings are supported by the positive experiences of international fans who visited South Africa during the event as highlighted in post-event research commissioned by FIFA.

Back in December 2008, FIFA commissioned a six-week study of South African residents with the aim of tracking public opinion towards the tournament from the initial build-up through to the final whistle and then beyond. The picture that emerged following the final wave of the survey is of a country that feels increasing pride in a tournament which was considered not only a huge success in its own right, but also an important event in terms of promoting national unity.

When asked in 2008 whether they thought the FIFA World Cup would be able to bring the South African people even closer together, 77 per cent of those asked said they believed this was a possibility. The post-event findings suggest that the event strengthened this sentiment, with 91 per cent of South Africans claiming their country is now more unified. The post-tournament results also showed an upswing in national confidence, with some 84 per cent feeling that their country had a stronger sense of self-belief post-tournament and 87 per cent feeling more confident than ever before in their nation's capabilities.

Sports
Africa
Economics
Diego Forlán
Octopus Paulo
.....

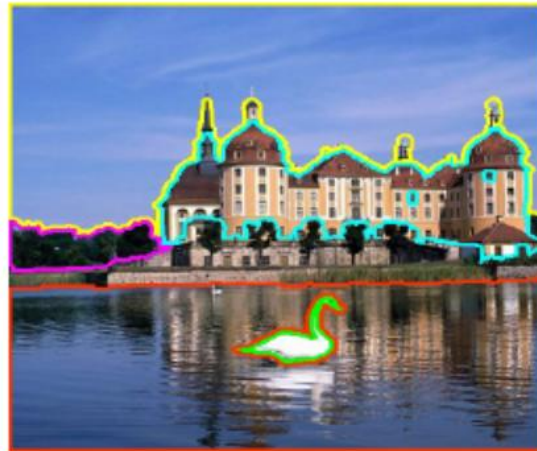
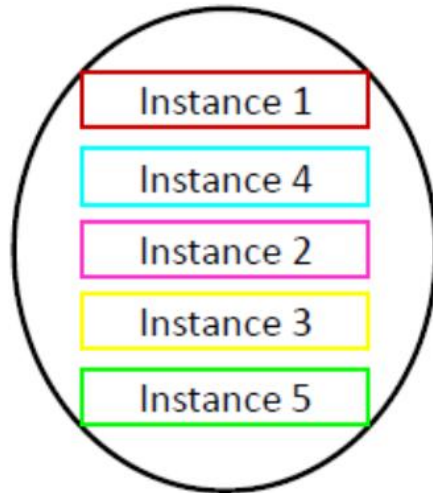
Multiple labels



$$X = \{X_1, X_2, X_3, \dots, X_N\}$$

bag---label

Multiple instance



sky
lake
building
swan
tree
...

<https://img-cdn.pixiv.org/>

Machine Learning: (3) Scenario

- The basic idea behind a **reinforcement learning** algorithm is this idea of what is called a **reward function**.



"Good dogs"



"Bad dogs"



- In reinforcement learning problems, you are usually asked to make a sequence of decisions over time.

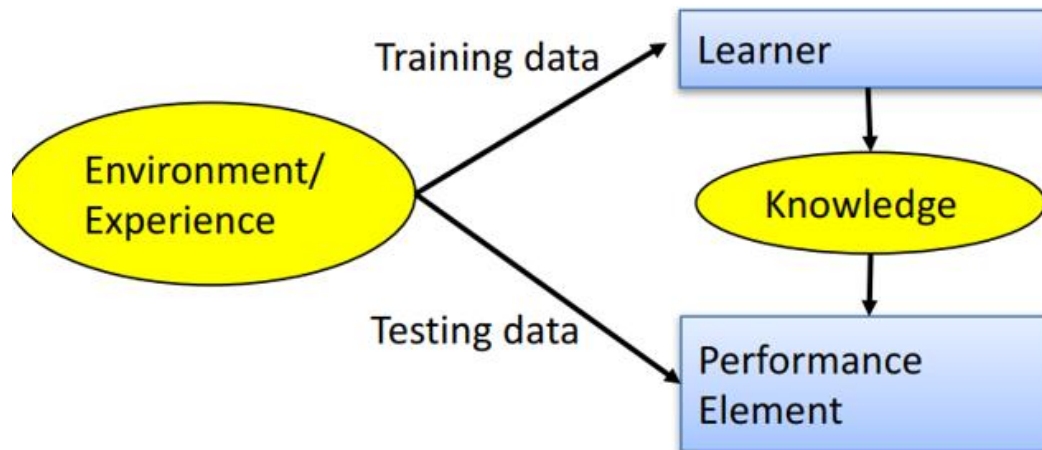
Reinforcement Learning



Framing a Learning Problem

- **Designing a Learning System**

- Choose the training experience
- Choose exactly what is to be learned
 - i.e. the **target function**
- Choose how to represent the target function
- Choose a learning algorithm to infer the target function from the experience



Framing a Learning Problem

- **Training vs. Test Distribution**

- We generally assume that the training and test examples are independently drawn from the same overall distribution of data
 - We call this “i.i.d” which stands for “independent and identically distributed”
- If examples are not independent, requires *collective classification*
- If test distribution is different, requires *transfer learning*

ML in a Nutshell

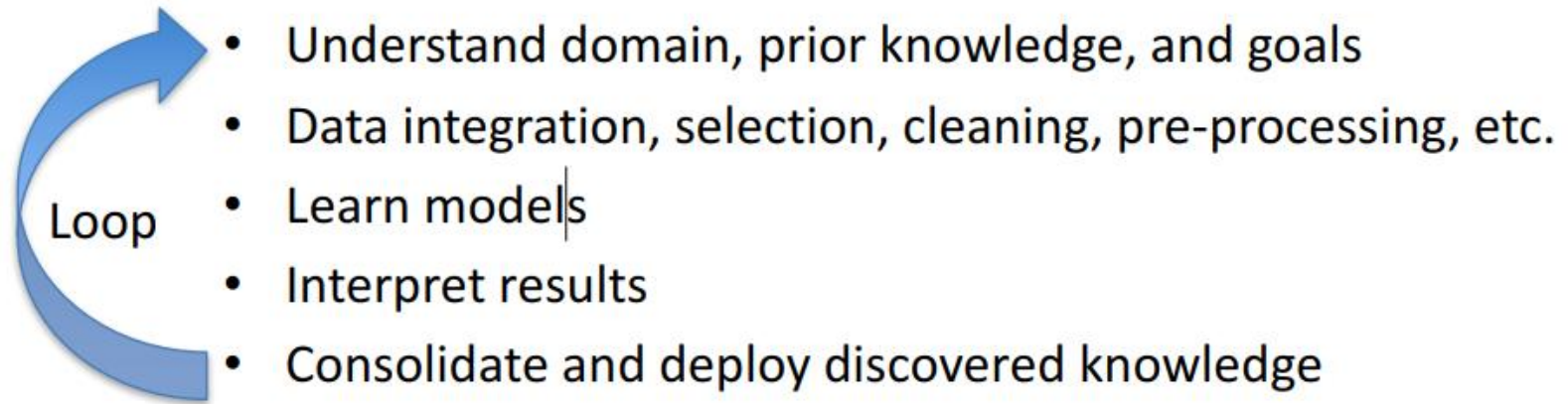
- Every ML algorithm has **three** components:

- Representation

- Optimization

- Evaluation

ML in Practice



Our focus will be on applying machine learning to real applications

History of Machine Learning

- 1950s

- –Samuel’ s checker player
- –Selfridge’ s Pandemonium

- 1960s

- Neural networks: Perceptron
- Pattern recognition
- Learning in the limit theory
- Minsky and Papert prove limitations of Perceptron

- 1970s

- Symbolic concept induction
- Winston’ s arch learner
- Expert systems and the knowledge acquisition bottleneck Quinlan’ s ID3
- Michalski’ s AQ and soybean diagnosis
- Scientific discovery with BACON
- Mathematical discovery with AM

History of Machine Learning

- 1980s:
 - –Advanced decision tree and rule learning
 - Explanation-based Learning (EBL)
 - Learning and planning and problem solving
 - Utility problem
 - Analogy
 - Cognitive architectures
 - Resurgence of neural networks (connectionism, backpropagation)Valiant' sPAC Learning Theory
 - Focus on experimental methodology
- 1990s
 - Data mining
 - Adaptive software agents and web applications Text learning
 - Reinforcement learning (RL)
 - Inductive Logic Programming (ILP)
 - Ensembles: Bagging, Boosting, and Stacking
 - Bayes Net learning

History of Machine Learning (cont.)

- 2000s

- Support vector machines & kernel methods
- Graphical models
- Statistical relational learning
- Transfer learning
- Sequence labeling
- Collective classification and structured outputs
- Computer Systems Applications (Compilers, Debugging, Graphics, Security)E-mail management
- Personalized assistants that learn
- Learning in robotics and vision

- 2010s

- Deep learning systems
- Learning for big data
- Bayesian methods
- Multi-task & lifelong learning
- Applications to vision, speech, social networks, learning to read, etc.???

Learning Theory

- Why the learning model work?
- When will the learning model work?
- How many labeled samples are needed to train a good model?
- How to evaluate the generalization capability of learning model?
-

Q.

- **What are the main tasks of machine learning?**
 - Classification, Regression
 - Recognition, Clustering
 - Computer vision, Speech recognition
 - Structural learning

Q.

- **When will we need machine learning?**
 - exists some " underlying pattern" to be learned
 - the problem cannot be programmable (easy) defined
 - data about the pattern is available
 - there are a large amount of data

Q.

- **Which of the following is best suited for categorizing machine learning:**
 - Shallow methods and deep methods
 - Linear methods and nonlinear methods
 - Empirical Risk Minimization(ERM) methods and Structural Risk Minimization(SRM) methods
 - Supervised methods and unsupervised methods

Q.

- **Which of the following is the scenario of machine learning?**
 - Supervised and unsupervised learning
 - Semi-supervised learning and transfer learning and other weakly supervised learning
 - Deep learning
 - Reinforcement learning